



Mini Project Report - 09

**Master of Computer Application – General
Semester – III**

Sub: Web Technologies

Topic: Authentication and Authorization

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August 2025

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INTRODUCTION

This mini project demonstrates a basic Role-Based Authorization System developed using Node.js and Express.js. The main objective of the project is to authenticate users based on their login credentials and authorize access to specific resources according to their roles.

The system supports three types of users: Admin, Faculty, and Student. After a successful login, a user is assigned a role. Certain operations, such as deleting student records, are restricted only to users with the Admin role. If a user without the required permission attempts to access the protected resource, the system displays an "Access Denied" message.

This project helps in understanding the concepts of authentication, authorization, HTTP requests, REST APIs, and role-based access control (RBAC) using Express.js. It also provides a foundation for building secure web applications where different users have different access permissions.

INPUT CODE

// AUTHORIZATION: Multile users with permission

```
const express = require('express')
const app = express();
app.use(express.json());
let currentRole = "";

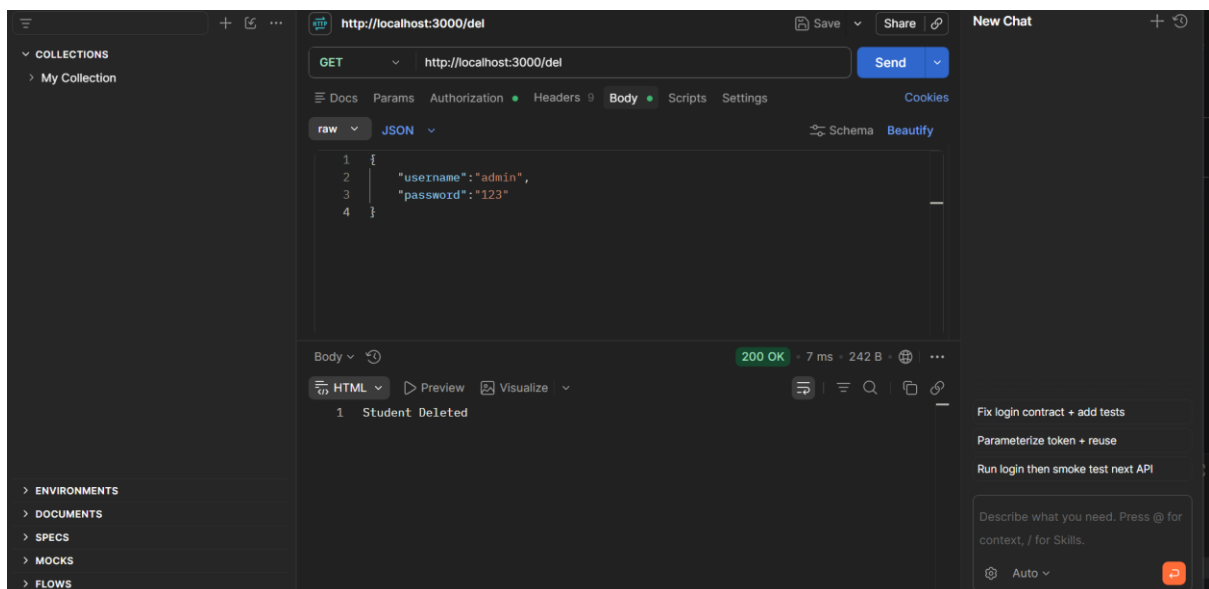
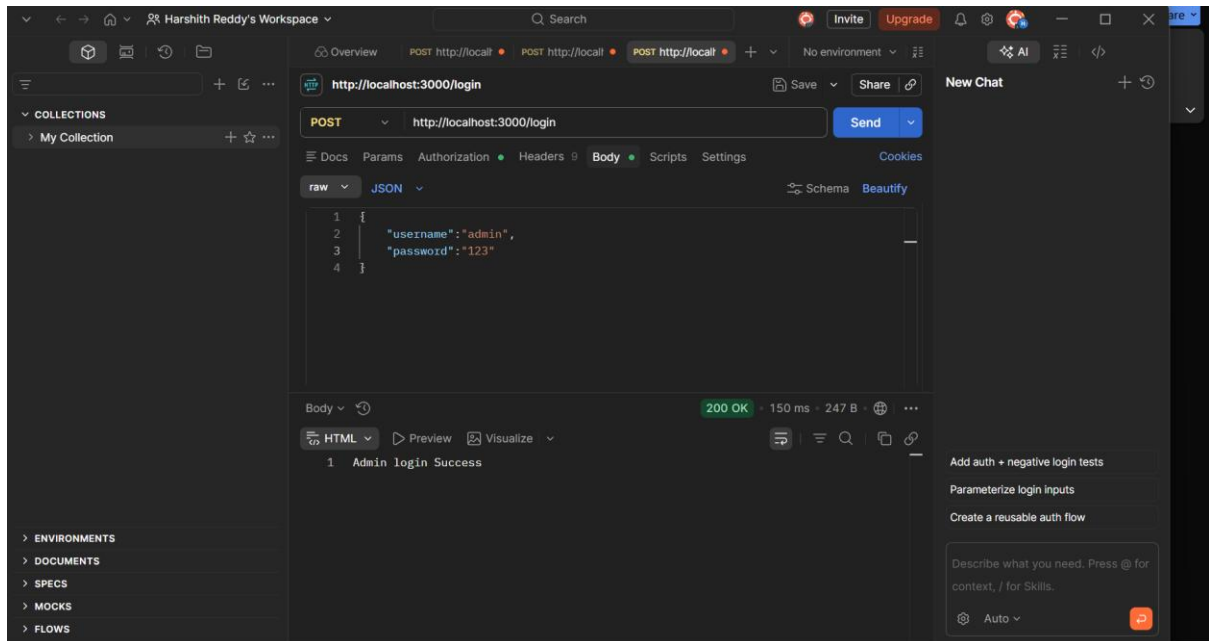
app.post("/login", (req,res)=>{
  const{username,password}= req.body;

  if(username === "admin" && password === "123"){
    res.send("Admin login Success")
    currentRole = "admin";
  }else if(username === "faculty" && password === "123"){
    res.send("faculty login Success")
    currentRole = "faculty"
  }else if(username ==="student" && password === "123"){
    res.send("Student login Success")
    currentRole = "student"
  }else{
    res.send("invalid credentials")
  }
});

app.get("/del",(req,res)=>{
  if(currentRole === "admin"){
    res.send("Student Deleted")
  }else{
    res.send("Access Denied??")
  }
});

app.listen(3000,()=>{
  console.log("http://localhost:3000/")
})
```

OUTPUT CODE



CONCLUSION

The Role-Based Authorization System successfully demonstrates the basic concepts of user authentication and authorization using Node.js and Express.js. The application verifies user credentials, assigns roles such as Admin, Faculty, and Student, and restricts access to protected resources based on the assigned role.

Through this project, we gained practical knowledge of building REST APIs, handling HTTP requests, implementing role-based access control (RBAC), and securing application endpoints. Although the project uses a simple in-memory role management approach, it provides a strong foundation for developing secure web applications.

In the future, the system can be enhanced by integrating a database, implementing JWT (JSON Web Token) authentication, encrypting passwords using bcrypt, supporting multiple concurrent users, and adding more user roles and permissions. These improvements would make the application more secure, scalable, and suitable for real-world use.